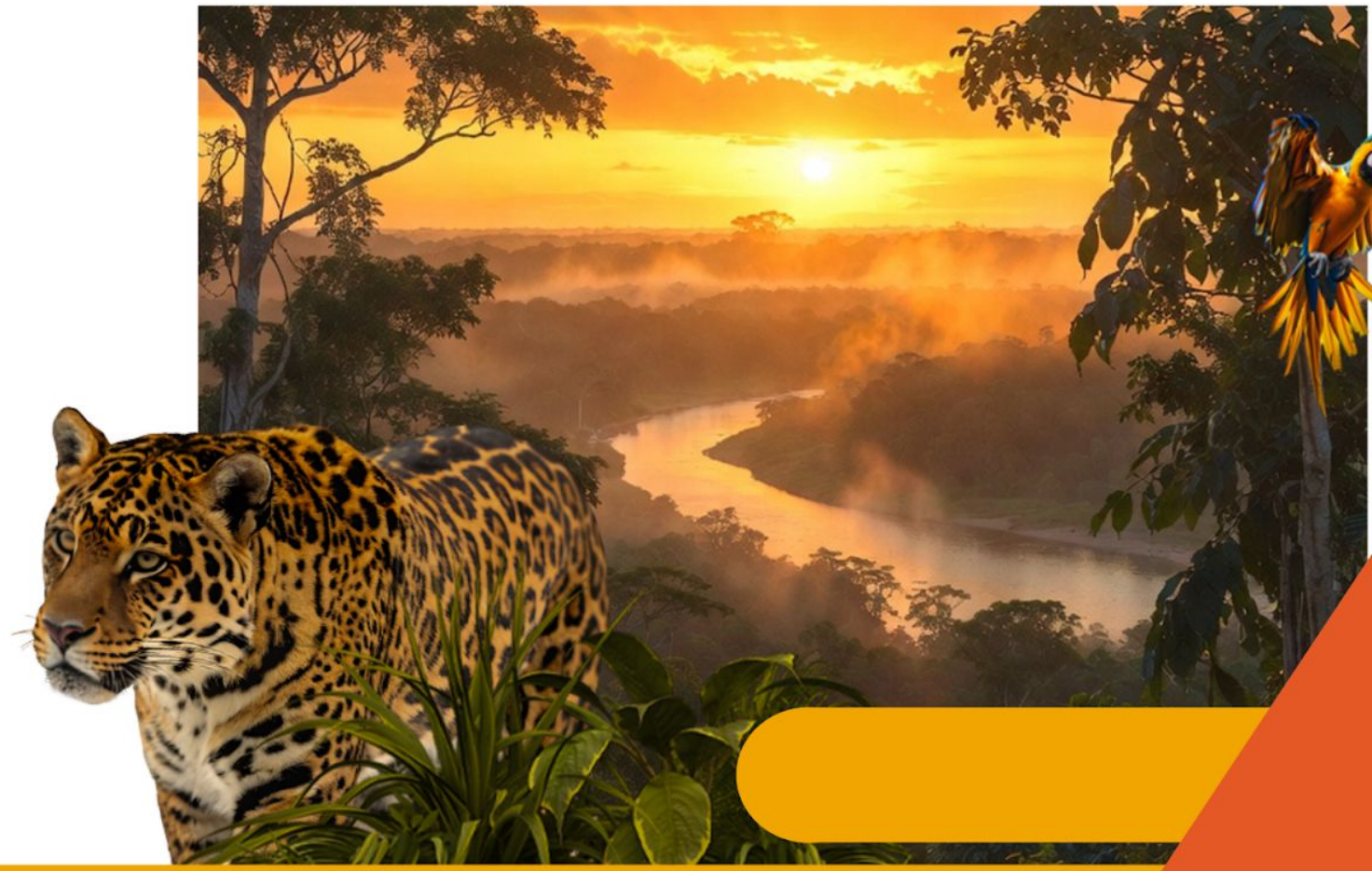
Smosaic

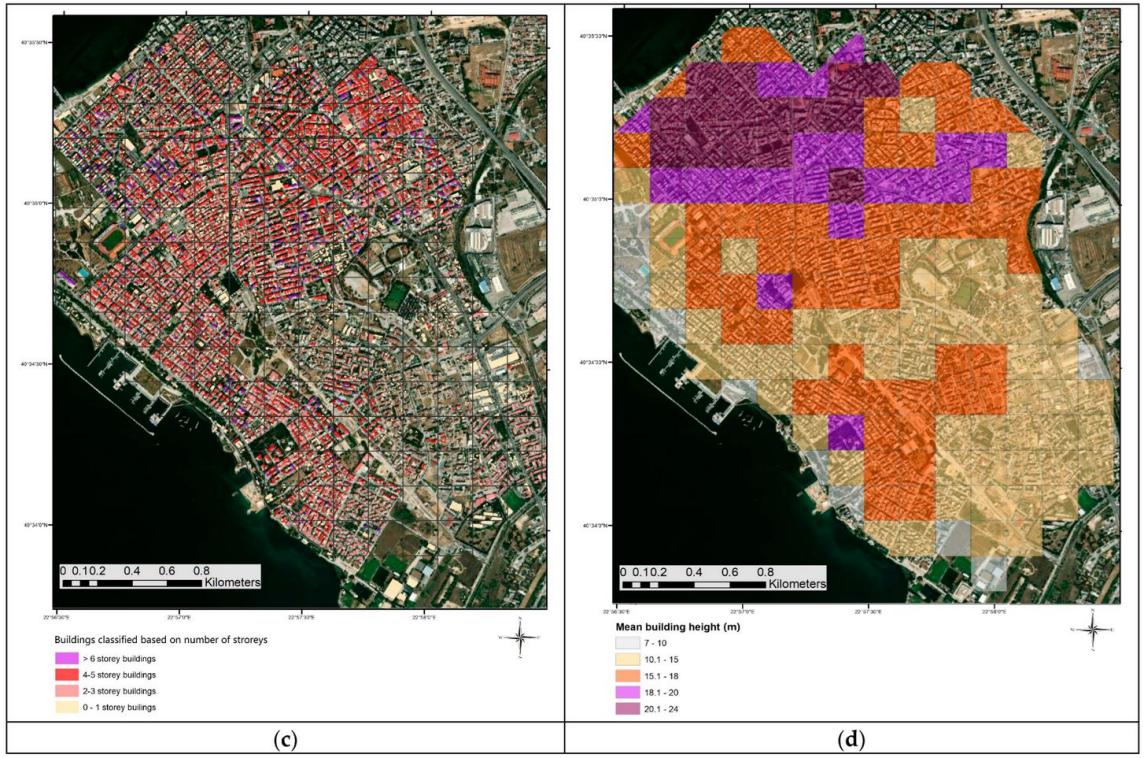
Gabriel Sansigolo

Brazilian National Institute for Space Research (INPE)



EO data

Earth Observation (EO) data includes orbital remote sensing data obtained by EO satellites and derived data obtained from EO satellites. These data allow us to broaden our understanding of the dimensions, location and evolution of targets over time.

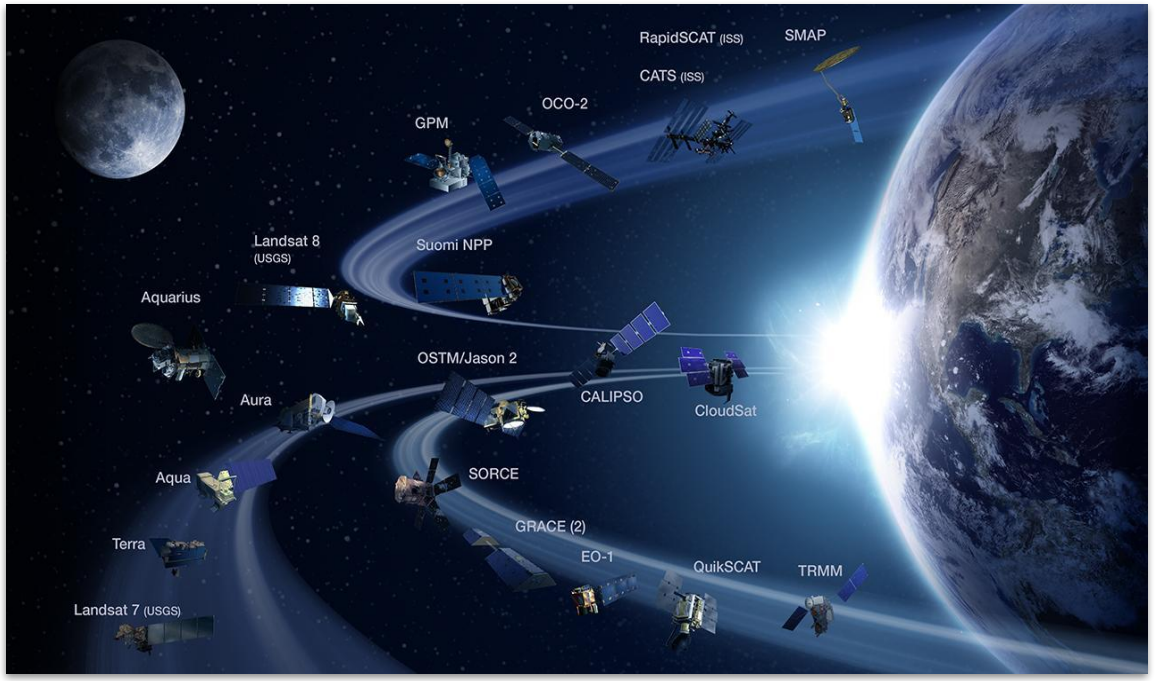


Source: (Stamou et al, 2025)

Big EO data

Satellites are constantly collecting a massive amount of EO data, from different satellites/sensors. When working with large volumes of EO, the enormous amount of data poses a significant challenge for analysts.

Systems need to update themselves to work properly in this “big EO data” era.



Source: NASA

Mosaic

Despite recent advances in accessing large volumes of EO data via services, there are still substantial challenges related to building seamless satellite mosaics using big EO data.

The generation of an image mosaic remains a complex process of merging multiple overlapping images into a single image.

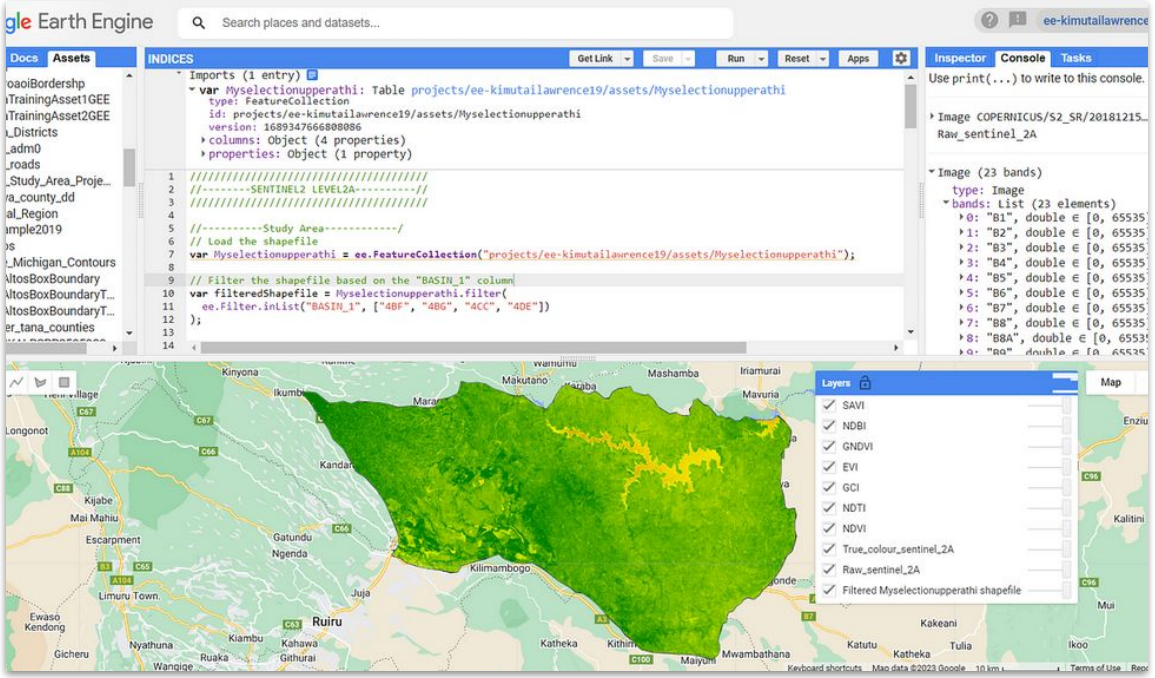


Source: Author

Google Earth Engine

Tools like Google Earth Engine (GEE) revolutionized geospatial analysis in the cloud. However, they present hidden barriers for governmental and long-term operations.

Restrictive terms of use; do not classify governmental operational use as "non-commercial". Limited by API rate limits and export bottlenecks.

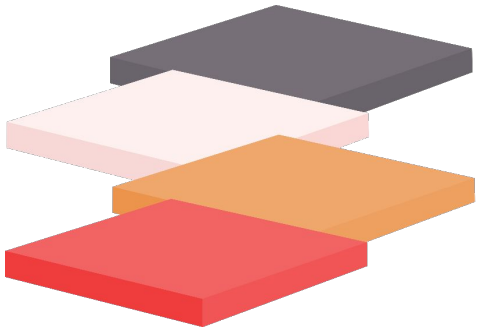


Source: GEE

Objective

Our work aims to develop a free and open-source tool for building seamless satellite image mosaics.

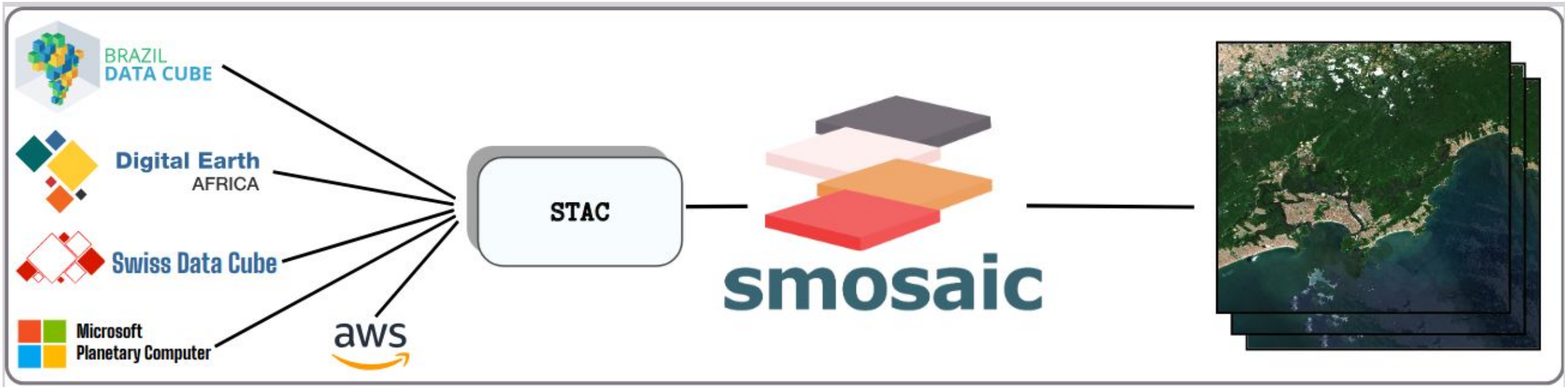
This tool was designed to be used by analysts without programming knowledge; therefore, remote sensing researchers, geoscience students, and analysts will be able to use it with no programming barriers.



smosaic

Source: Author

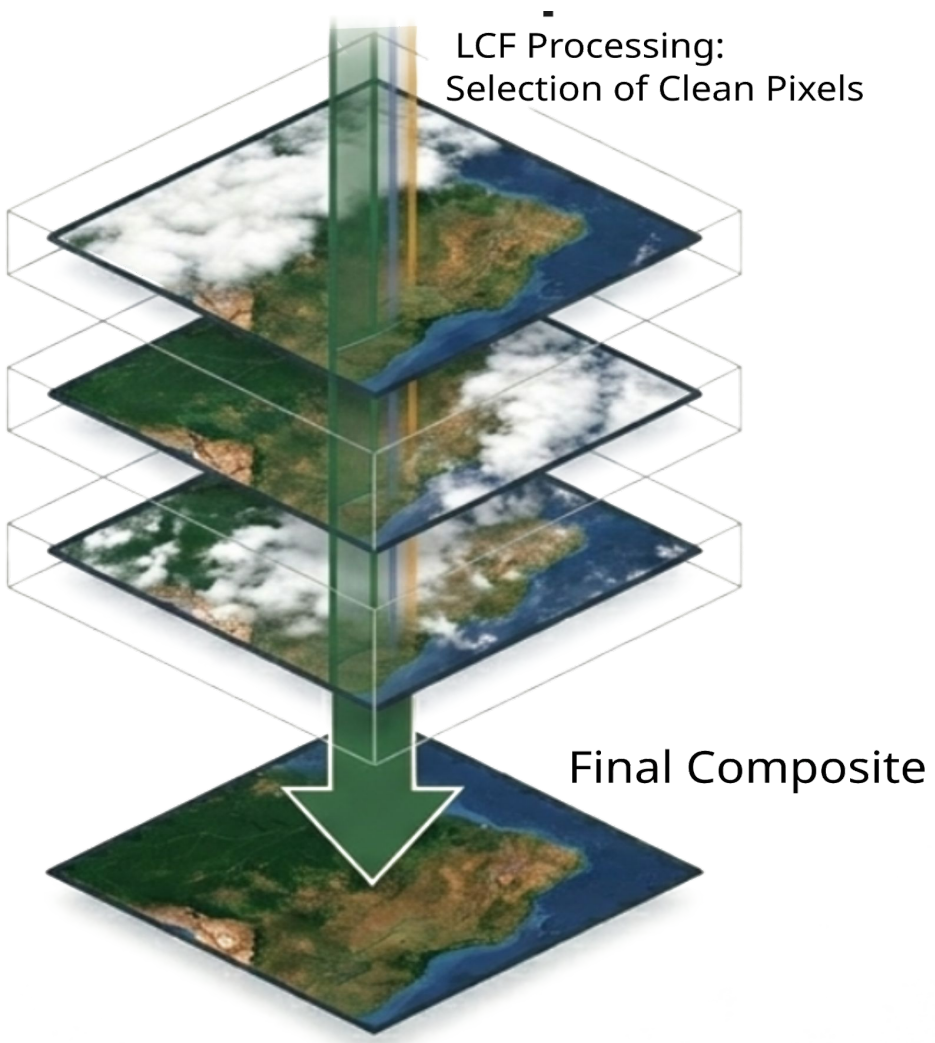
Architecture



Compositing functions

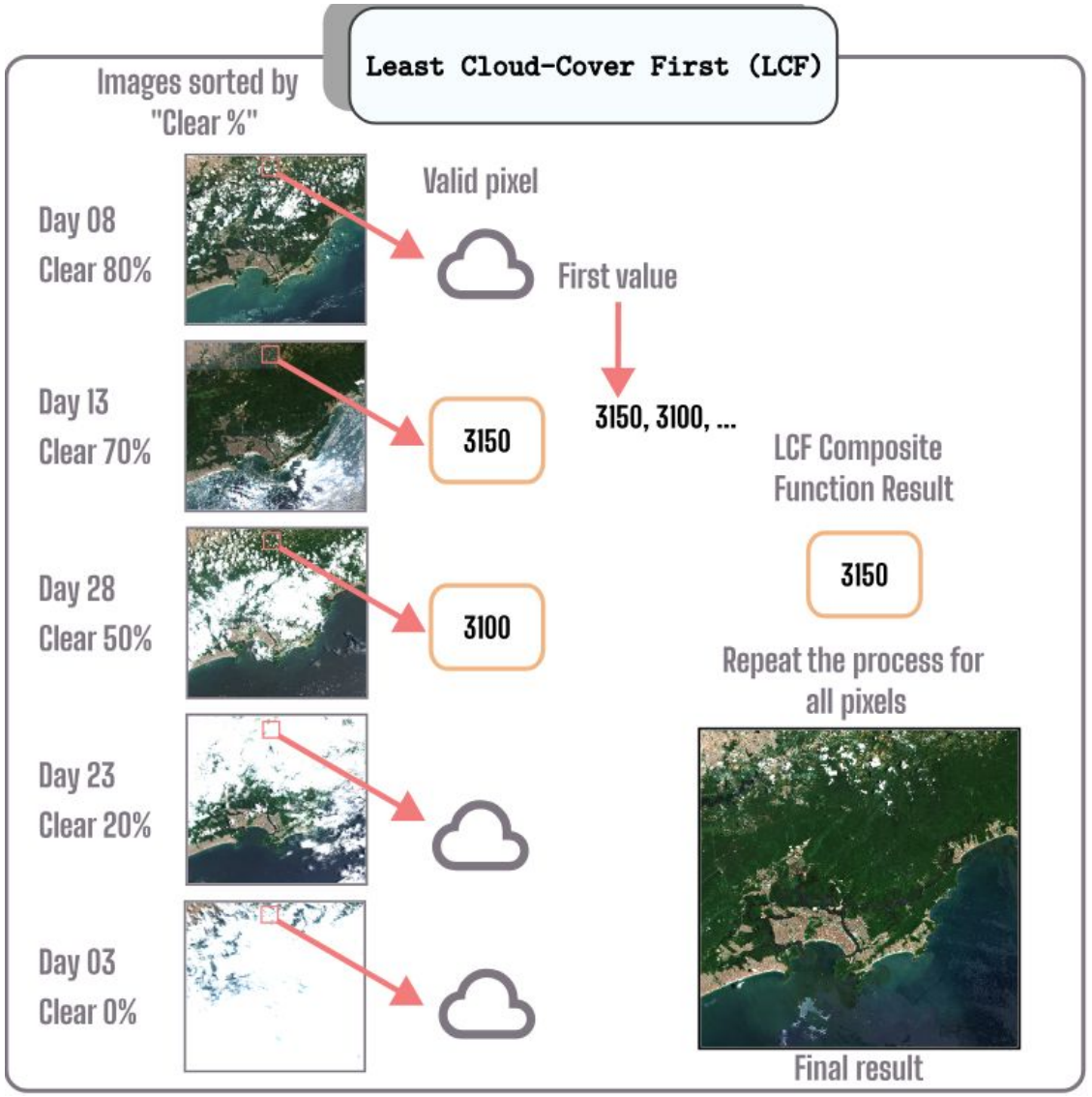
The process_period() function orders and clips all downloaded images based on the **LCF** (least cloud-cover first), **CTD** (closest to date), or **chrono** (chrono-logical) compositing functions, performing ordering and clipping.

The tool also attaches a provenance band, with the exact acquisition date of each individual pixel used in the final mosaic.



Source: Author

LCF


Least Cloud-Cover First (LCF)
Images sorted by "Clear %"
Valid pixel
Day 08
Clear 80%
First value
Day 13
Clear 70%
3150
3150, 3100, ...
LCF Composite Function Result
Day 28
Clear 50%
3100
3150
Repeat the process for all pixels
Day 23
Clear 20%
Day 03
Clear 0%
Final result

Mosaic function

```python
import os
from smosaic import mosaic

stac_url="https://data.inpe.br/bdc/stac/v1"

result = mosaic(
    name="para",
    data_dir=os.path.abspath(""),
    stac_url=stac_url,
    collection="S2_L2A-1",
    bbox="-53.5007,-2.0485,-52.5943,-1.1288",
    output_dir=os.path.join("output"),
    mosaic_method="lcf",
    start_year=2025,
    start_month=1,
    start_day=1,
    duration_days=16,
    bands=["B02","B03","B04"]
)
```

Experiment

To demonstrate the tool's potential, we applied smosaic to the BDC-Favelas project (Center for Favela Studies - CEFAVELA).

Brazilian favelas lack systematic and open data about their expansion, size, and changes over time.

Produce time series of images and spectral indices for the ABC Paulista region throughout the entire year of 2024, using images from the Sentinel-2 collection.



Source: Author

Urban Analysis

Target:
ABC Region

Parameter:
Profile=”urban_analysis

Output:
108 Cloud-Optimized GeoTIFFs
(12 mo x 6 products)



Bands:
RGB, NIR, SWIR, SLC, PROVENANCE

NDBI:
Normalized Difference Built-up Index

NDVI:
Normalized difference vegetation index

Code availability

PyPI smosaic:
https://pypi.org/project/smosaic/

Github smosaic:
https://github.com/brazil-data-cube/smosaic

Github Paper Compedium:
https://github.com/GSansigolo/gee-image-mosaic-script

BDC-Favelas Explorer:
https://geolab.inpe.br/bdc/favelas/explorer

Read the Docs smosaic:
https://smosaic.readthedocs.io

Thank you!

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